In [1]:
```
!pip install gensim scikit-learn
```

```
Requirement already satisfied: gensim in C:\Users\manis\AppData\Local\Python\pyth
oncore-3.12-64\Lib\site-packages (4.4.0)
Requirement already satisfied: scikit-learn in C:\Users\manis\AppData\Local\Pytho
n\pythoncore-3.12-64\Lib\site-packages (1.9.1)
Requirement already satisfied: numpy>=1.18.5 in C:\Users\manis\AppData\Local\Pyth
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n\pythoncore-3.12-64\Lib\site-packages (from gensim) (1.18.1)
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Requirement already satisfied: joblib>=1.4.0 in C:\Users\manis\AppData\Local\Pyth
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Requirement already satisfied: narwhals>=2.0.1 in C:\Users\manis\AppData\Local\Py
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Requirement already satisfied: threadpoolctl>=3.5.0 in C:\Users\manis\AppData\Loc
al\Python\pythoncore-3.12-64\Lib\site-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: cloudpickle>=3.0 in C:\Users\manis\AppData\Local\P
ython\pythoncore-3.12-64\Lib\site-packages (from joblib>=1.4.0->scikit-learn) (3.
1.2)
Requirement already satisfied: wrapt in C:\Users\manis\AppData\Local\Python\pytho
ncore-3.12-64\Lib\site-packages (from smart_open>=1.8.1->gensim) (2.5.2)
```

In [7]:
```
# 1. Import necessary packages
import pandas as pd
import numpy as np
import re
from gensim.models import Word2Vec
from sklearn.metrics.pairwise import cosine_similarity

# 2. Load Twitter dataset
data = pd.read_csv('twitter_dataset.csv')

# Select the correct Text column
tweets = data['Text'].dropna().astype(str)

# 3. Clean and tokenize tweets
def clean_text(text):
    text = text.lower()
    text = re.sub(r'http\S+|www\S+', '', text)
    text = re.sub(r'[^a-z\s]', '', text)
    return text.split()

sentences = [clean_text(tweet) for tweet in tweets]
sentences = [s for s in sentences if len(s) > 0]

# 4. Train CBOW Word2Vec model
model = Word2Vec(
    sentences=sentences,
    vector_size=100,
    window=5,
    min_count=1,
    sg=0,
    epochs=10
)

print("CBOW Model Training Completed!")
```

```
# 5. Convert tweets into document vectors
def get_vector(words):
    vectors = [model.wv[w] for w in words if w in model.wv]

    if len(vectors) == 0:
        return None

    return np.mean(vectors, axis=0)

document_vectors = np.array([
    get_vector(s) for s in sentences
])

# 6. Take user input
query = input("Enter a topic or tweet: ")

query_words = clean_text(query)
query_vector = get_vector(query_words)

# 7. Calculate similarity and recommend tweets
if query_vector is None:
    print("No matching words found in vocabulary. Try another topic.")

else:
    similarity = cosine_similarity(
        [query_vector], document_vectors
    )[0]

    top_indices = np.argsort(similarity)[::-1][:5]

    print("\nRecommended Similar Tweets:")

    for i in top_indices:
        print("\nTweet:", tweets.iloc[i])
        print("Similarity:", round(similarity[i], 3))

# 8. Save the model
model.save('recommendation_cbow.model')

print("\nRecommendation Completed!")
```

```
CBOW Model Training Completed!
```

```
Recommended Similar Tweets:

Tweet: Quickly international show mission thank necessary. Can put minute party a
ble.
Fact series politics team alone we open strategy. Approach arrive prove idea.
Similarity: 0.985

Tweet: Model management least miss defense particular trouble. Try article tell c
are summer site.
Indicate grow behavior other language. Occur relationship mention consider party
table performance so.
Gas out adult stuff.
Similarity: 0.984

Tweet: Step measure there. Event tree truth white. Again set country address.
Accept student individual build. Civil shoulder group alone party.
Area day indicate especially. Lay lead administration sense trip traditional. Pol
itics center down street strong.
Similarity: 0.984

Tweet: Part among chair several party performance stay. Mind choose perhaps natur
e section last. Two consider unit various ball future.
Art record degree theory. Treatment view through long kind point focus party. War
south act name her.
Similarity: 0.984

Tweet: Investment guy whose source. Television person choice each. Education inte
resting natural specific.
Character rock movement detail six shake party. Blood generation travel country c
andidate by themselves.
Push democratic head room trip air.
Similarity: 0.984

Recommendation Completed!
```

In [ ]: